Date: 12 / 03 / 2026

Duration: 1 hour 45

CONTINUOUS ASSESSMENT 1

III YEAR – SEMESTER 6

CSE23AE304 - PCP V

(B.Tech. E01,E02,E03,E04)

SET F

Dataset F – Academic

Each student object contains:

- id
- name
- gender
- marks → { dsa, mern, aptitude }
- attendance

All endpoints must be implemented under the base route: /students

Proper RESTful design, status codes, and error handling are mandatory.

Question 1 – Highest Total Marks

GET /students/topper

Return the student who has the highest total marks.

Total marks = dsa + mern + aptitude

Requirements

- Must calculate total dynamically.
- Must use reduce or sorting.
- Include totalMarks in response.
- Must not modify original dataset.

Example Response

```
{
 "id": 37,
 "name": "Student37",
 "gender": "Female",
 "totalMarks": 285
}
```

Question 2 – Get Student By ID

GET /students/:id

Return student details based on ID.

Requirements

- Use route parameter.
- ID must be validated.
- If student not found → return 404.
- Server must not crash for invalid input.

Example Response

```
{
 "id": 15,
 "name": "Student15",
 "gender": "Female",
 "marks": {
  "dsa": 85,
  "mern": 88,
  "aptitude": 82
 },
 "attendance": 89
}
```

Example Error Response

```
{
 "message": "Student not found"
}
```

Question 3 – Filter by MERN marks

GET /students/filter?minMern=VALUE

Return students whose MERN marks are greater than or equal to the given value.

Requirements

- Use query parameter.
- Access nested marks.mern.
- If query parameter missing → return 400.
- Return filtered list only.

Example Request

GET /students/filter?minMern=90

Example Response

```
[
 {
  "id": 2,
  "name": "Student2",
  "gender": "Female",
  "marks": {
   "dsa": 92,
   "mern": 95,
```

```
    "aptitude": 89
   },
   "attendance": 96
  },
  {
   "id": 13,
   "name": "Student13",
   "gender": "Female",
   "marks": {
    "dsa": 93,
    "mern": 97,
    "aptitude": 90
   },
   "attendance": 98
  }
]
```


Example Error Response


```
{
  "message": "minMern query parameter is required"
}
```


Question 4 – Gender-wise Top Scorer

GET /students/top/:gender

Return the highest total scorer among a specific gender.

Requirements

- Use route parameter.
- Gender comparison must be case-insensitive.
- Must compute total marks dynamically.
- If invalid gender → return 400.

Example Request

GET /students/top/female

Example Response


```
{
  "gender": "Female",
  "id": 37,
  "name": "Student37",
  "totalMarks": 285
}
```


Example Error Response


```
{
  "message": "Invalid gender provided"
}
```


Question 5 – Attendance Analysis

GET /students/attendance/average

Return:

- Average attendance
- Number of students with attendance above 90%

Requirements

- Must use reduce.
- Average must be rounded to two decimal places.
- Return structured JSON response.

Example Response


```
{
  "averageAttendance": 88.42,
  "above90Count": 14
}
```